

EXP 2-2 Experimental Plan M9

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I Aim of the Experiment

The objective of this experiment is to determine the shear modulus (G) of metal and polymer wires by analyzing their oscillatory motion in a torsion pendulum setup. The experiment will also assess measurement uncertainties and their effect on the final calculated values.

II Principle of the Experiment

A torsion pendulum consists of a wire under torsional stress that induces simple harmonic motion. The shear modulus (G) of the wire material can be determined using the following equation:

$$G = \frac{4\pi LmR^2}{r^4T^2} \quad (1)$$

Where L is the length of the wire, r is the radius of the wire, m is the mass of the disk, R is the radius of the disk, and T is the period of oscillation.

For polymers, internal energy loss is not negligible, resulting in internal damping, as follows:

$$\text{The quality factor, } Q = \frac{\pi}{\kappa}$$

$$\kappa = \frac{\ln \frac{A_0}{A_n}}{n}$$

III Measurement Procedure

- 1) Calibrate the device.
- 2) Gauge the disc's diameter.
- 3) Weigh a disc using a scale.
- 4) Measure the wire's diameter.
- 5) Measure the polymer wire's length, assemble the setup vertically, twist it, and record the twist angle, oscillation period, and amplitude.
- 6) Measure the metal wire's length, assemble the setup vertically, twist it, and record the torsion angle, oscillation period, and amplitude.

- 7) Create a large angular displacement of the disc (as the angle diminishes with each loop), release it to initiate oscillation, measure the time for 10 loops with a timer, and then calculate the period for one oscillation.

IV Estimating L and T

The relationship between shear modulus and Young's modulus (E) is:

$$G = \frac{E}{2(1 + \nu)}$$

Where ν is Poisson's ratio.

The calculation shows that the G is $80.77 \times 10^9 \text{ Pa}$ for metal wire and $60.71 \times 10^7 \text{ Pa}$ for polymer wire.

According to the formular (1), the relationship between T and L is derived as:

$$T^2 = \frac{4\pi m R^2}{r^4 G} L \quad (2)$$

For each line, the team selected six lengths equidistant and brought them into the formula in (2) to draw the image of T^2 -L, as follows:

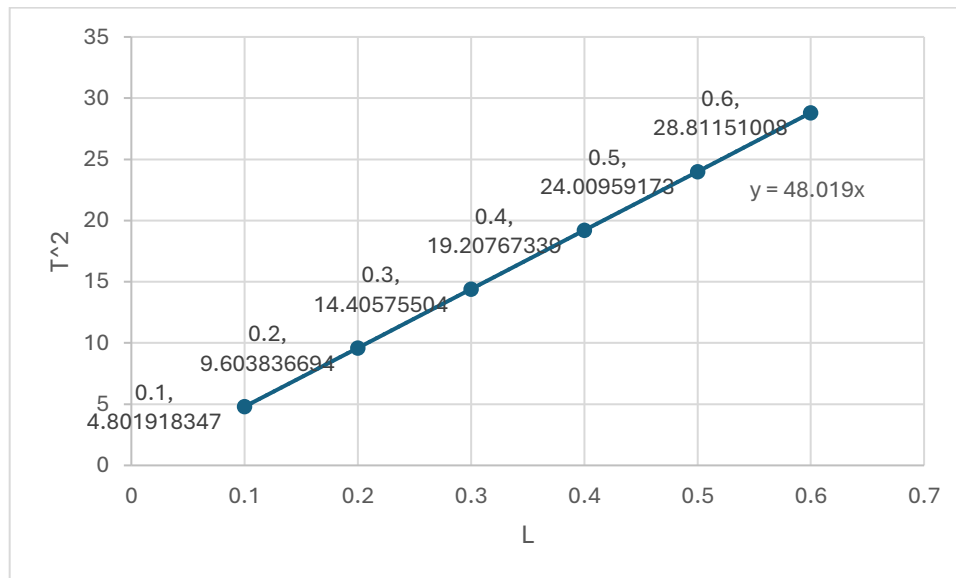


Figure 1. The T^2 -L prediction graph of metal wire.

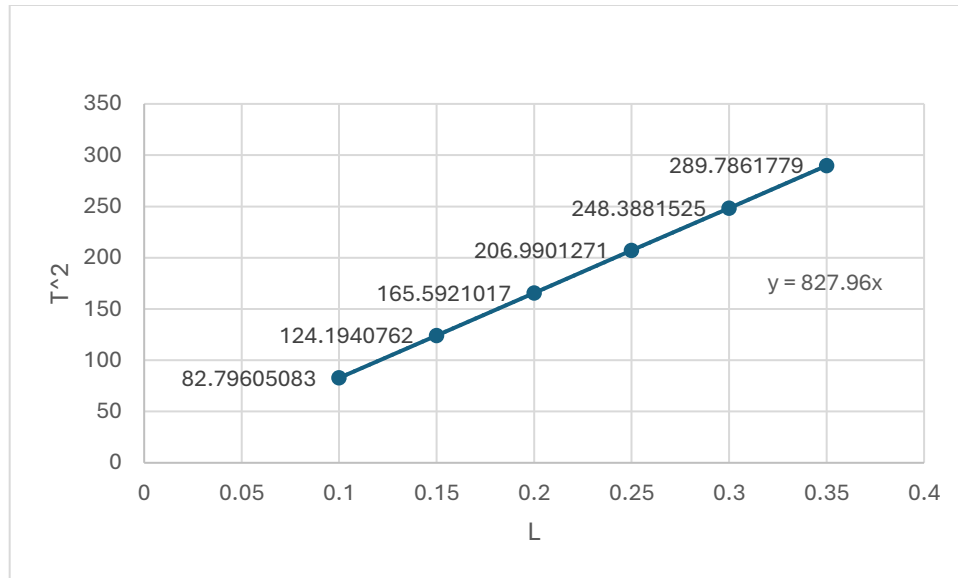


Figure 2. The T^2 -L prediction graph of polymer wire

V The Values to be Measured

The selection of L between 10 - 60 cm and 10 - 30 cm is because the longer the wires, the shorter the period. Considering the limited experimental time, it is necessary to rationally control the length of the wires to ensure that all data can be measured within the available time. And the difference in values between metal and polymer wires lies in their period difference. As metal wire has a shorter period than polymer one, the length of metal wire is set slightly longer than that of polymer wire during value assignment. All the values to be measured are shown in Table 1.

Table 1. All the estimated values and parameters that need to be measured.

	Meaning	Approx value	Instrument	Resolution	Precision	Number of measurements
L	Length of the metal wire	10cm-60cm	Ruler	1mm	1%	3 times each length
	Length of the polymer wire	10cm-30cm				
m	Mass of the disc	1000g	Balance	1g	0.1%	3
R	Disc radius	50mm	Ruler	1mm	2%	3 times in 3 directions
r	Matel wire radius	0.3mm	Micrometer	0.01mm	3.3%	3 times in 3 directions
	Polymer wire radius	0.5mm			2%	
T	Time of one oscillation cycle	2s-20s	Stopwatch	0.01s	0.5%	3 times 5 cycles for each period
				0.2s(human)		

A_0	Angle of twist at the beginning	180degrees	Protractor	1degree	0.55%	3 times
A_n	Maximum angle of the n^{th} cycle	90-180 degrees	Protractor	1degree 5 degrees(human)	0.55%	5 data points for 1 cycle